```
Calibration results
====================
Normalized Residuals
----------------------------
Reprojection error (cam0):     mean 0.16967374428136012, median 0.13438689597283598, std: 0.13313577764320805
Reprojection error (cam1):     mean 0.16678221728197207, median 0.13106020743972185, std: 0.1310000717207641
Gyroscope error (imu0):        mean 0.26530698561668203, median 0.24271255692239904, std: 0.18329182864805785
Accelerometer error (imu0):    mean 0.16058256233840704, median 0.13238405607587203, std: 0.1433738901615113

Residuals
----------------------------
Reprojection error (cam0) [px]:     mean 0.16967374428136012, median 0.13438689597283598, std:
0.13313577764320805
Reprojection error (cam1) [px]:     mean 0.16678221728197207, median 0.13106020743972185, std:
0.1310000717207641
Gyroscope error (imu0) [rad/s]:     mean 0.0016282515319009673, median 0.0014895841950859198, std:
0.0011249051738589583
Accelerometer error (imu0) [m/s^2]: mean 0.022040489972957576, median 0.01817015134165566, std:
0.019678542566965357

Transformation (cam0):
------------------------
T_ci:  (imu0 to cam0):
[[ 0.03091207 -0.99952176  0.00082836  0.01295164]
[-0.01853232 -0.00140176 -0.99982728  0.03107439]
[ 0.99935029  0.03089138 -0.01856679 -0.09131488]
[ 0.          0.          0.          1.        ]]

T_ic:  (cam0 to imu0):
[[ 0.03091207 -0.01853232  0.99935029  0.09143107]
[-0.99952176 -0.00140176  0.03089138  0.01580985]
[ 0.00082836 -0.99982728 -0.01856679  0.02936287]
[ 0.          0.          0.          1.        ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.008955888179582315


Transformation (cam1):
```

```
-----------------------
T_ci:  (imu0 to cam1):
[[ 0.03040429 -0.99953755  0.00050747 -0.03703754]
[-0.02070746 -0.00113749 -0.99978493  0.03125783]
[ 0.99932316  0.03038724 -0.02073247 -0.09100958]
[ 0.          0.          0.          1.        ]]

T_ic:  (cam1 to imu0):
[[ 0.03040429 -0.02070746  0.99932316  0.09272135]
[-0.99953755 -0.00113749  0.03038724 -0.03421933]
[ 0.00050747 -0.99978493 -0.02073247  0.02938305]
[ 0.          0.          0.          1.        ]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
-0.008973028364751612

Baselines:
----------
Baseline (cam0 to cam1):
[[ 0.99999982  0.00033026 -0.00050198 -0.05004528]
[-0.00033135  0.9999976  -0.00216635 -0.00001002]
[ 0.00050126  0.00216651  0.99999753  0.00023126]
[ 0.          0.          0.          1.        ]]
baseline norm:  0.050045814036265006 [m]


Gravity vector in target coords: [m/s^2]
[-0.17582747 -9.80494738 -0.02268074]


Calibration configuration
=========================

cam0
-----
```

Camera model: pinhole
Focal length: [387.26496788634023, 388.0178836701606]
Principal point: [320.53538193017073, 238.7961117373796]
Distortion model: radtan
Distortion coefficients: [-0.0048964025986568405, -0.0032256738628326904, 6.867803461190135e-05, 0.0003187277311499132]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.088 [m]
Spacing 0.026399999999999996 [m]

cam1

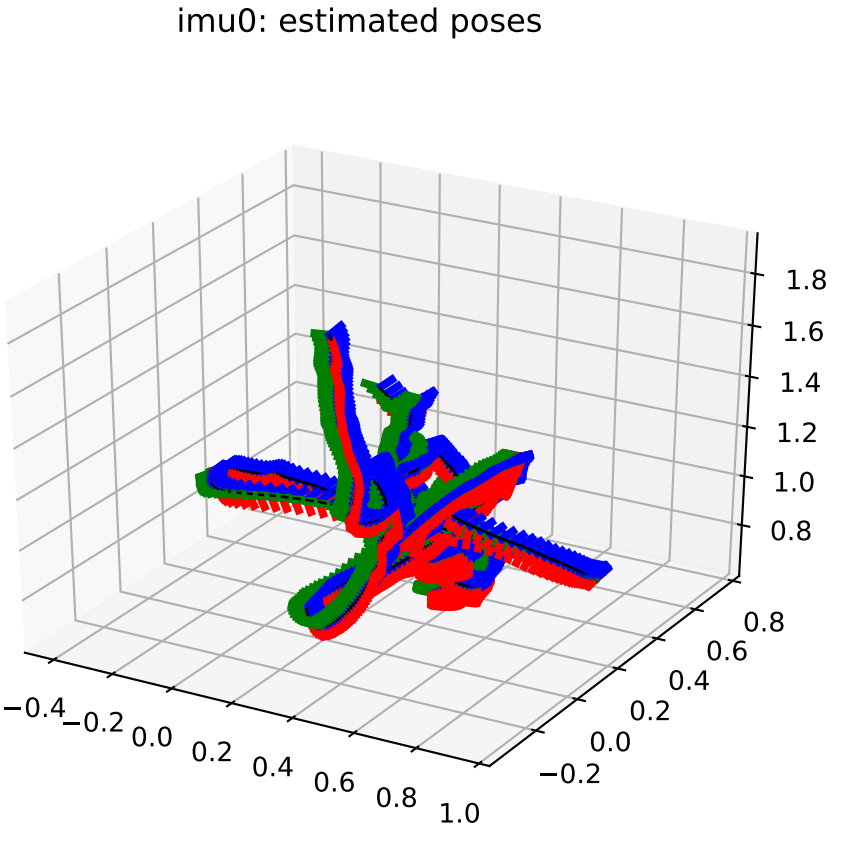
Camera model: pinhole
Focal length: [388.18861658777746, 388.9712538925426]
Principal point: [321.2136971603922, 238.95658293993972]
Distortion model: radtan
Distortion coefficients: [-0.002870215094047284, -0.00471202846643744, 5.440321557997829e-05, -9.458602566352592e-05]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.088 [m]
Spacing 0.026399999999999996 [m]

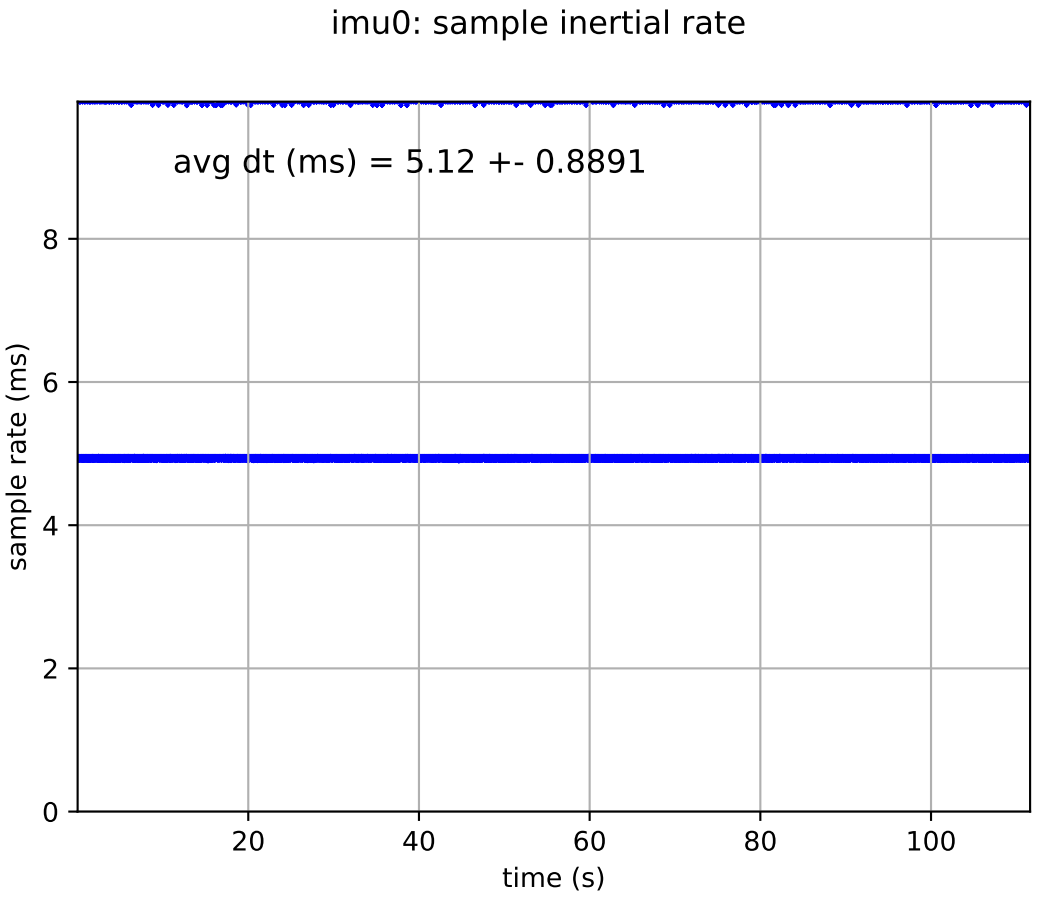
IMU configuration
==========

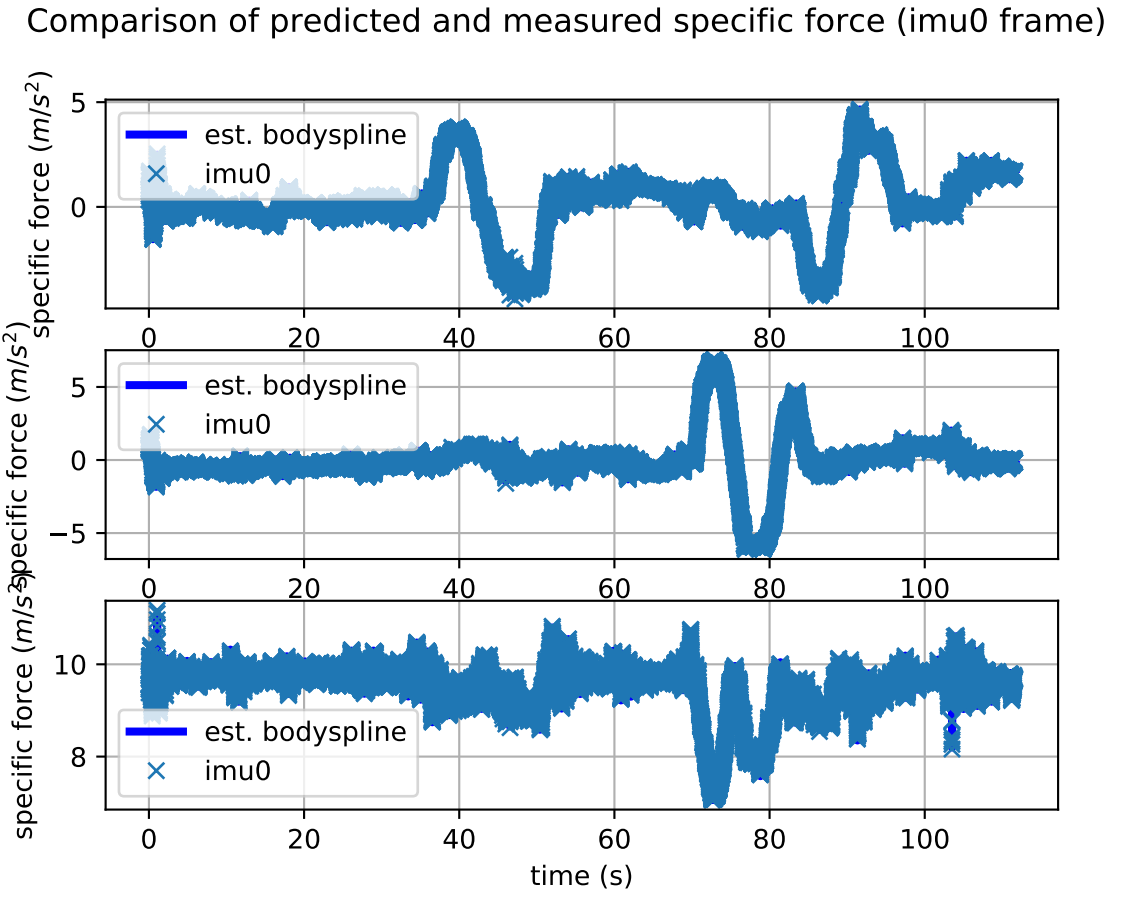
IMU0:

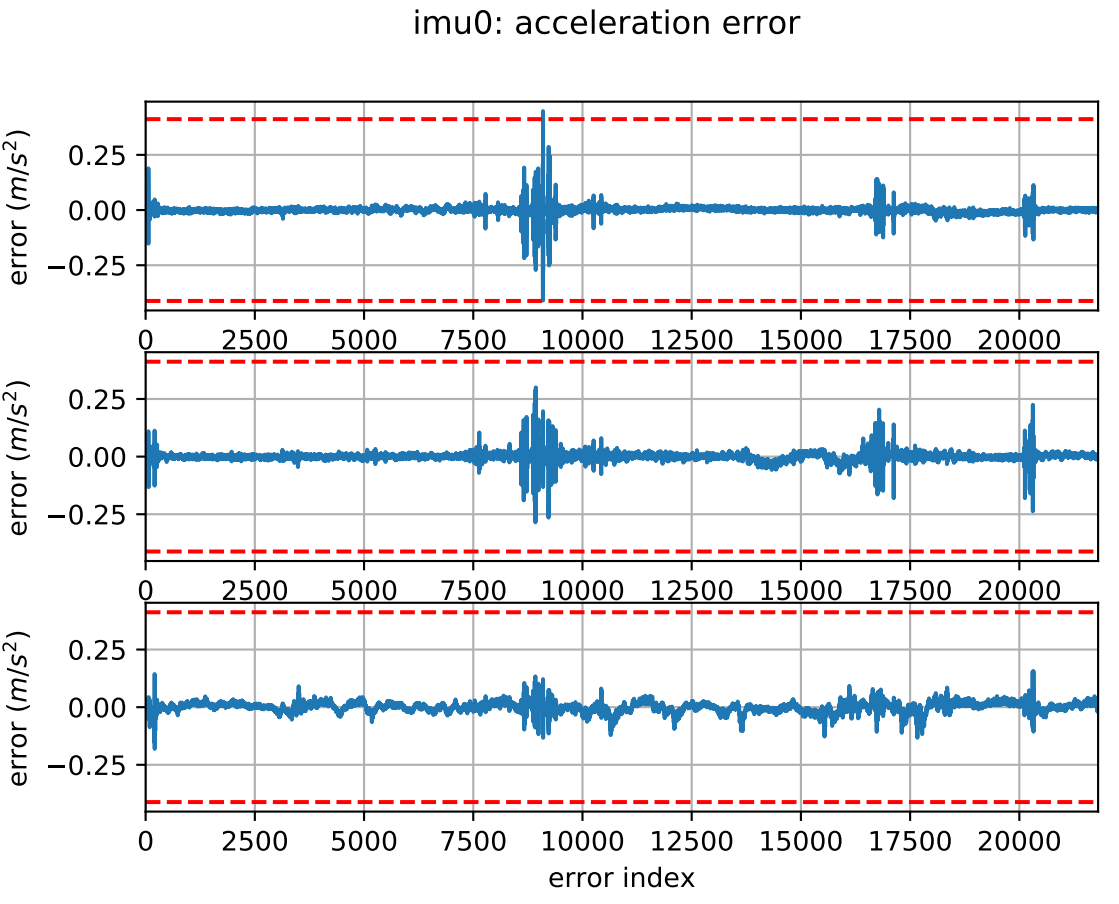
Model: calibrated
Update rate: 195.358538

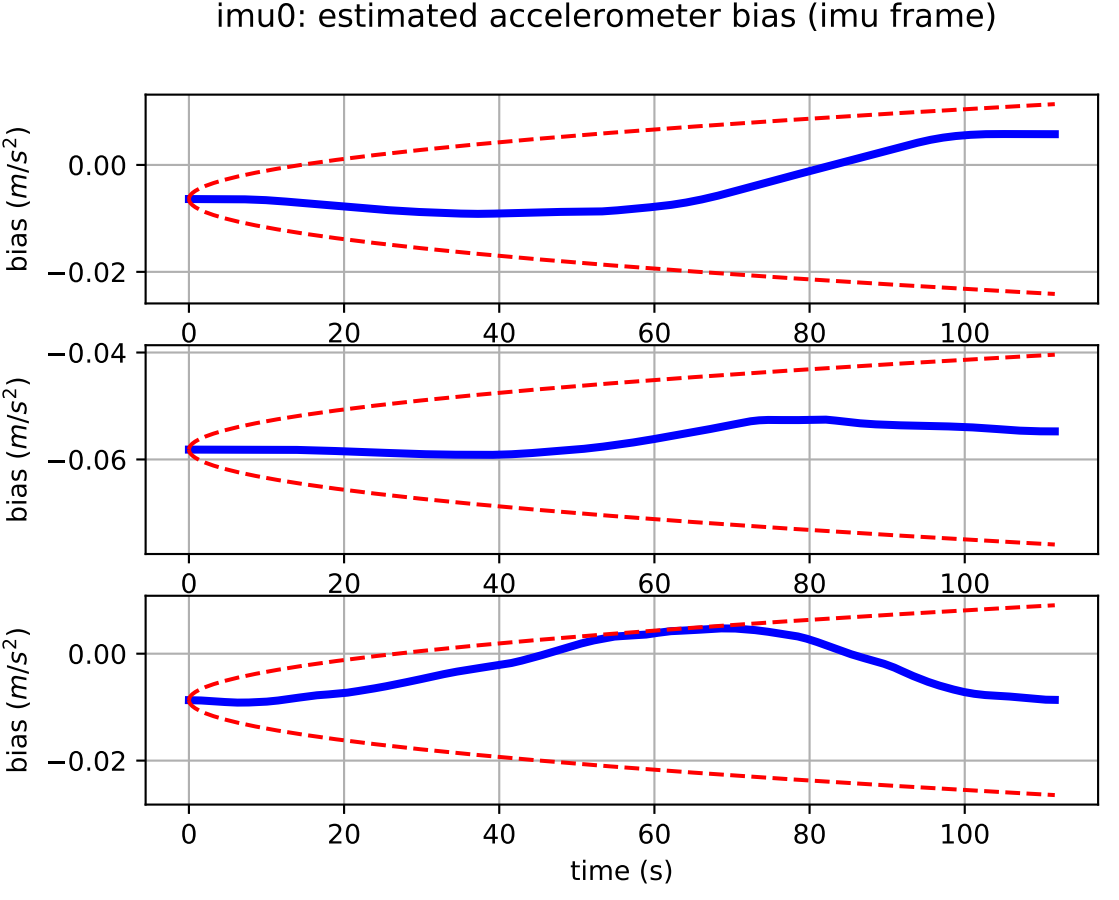
```
Accelerometer:
Noise density: 0.00981989090996197
Noise density (discrete): 0.13725332098332124
Random walk: 0.0005601293637217548
Gyroscope:
Noise density: 0.00043909309937990626
Noise density (discrete): 0.00613723580672497
Random walk: 1.6907369481278766e-05
T_ib (imu0 to imu0)
[[1. 0. 0. 0.]
[0. 1. 0. 0.]
[0. 0. 1. 0.]
[0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]
```


imu0: estimated poses
1.8
1.6
1.4
1.2
1.0
0.8
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0

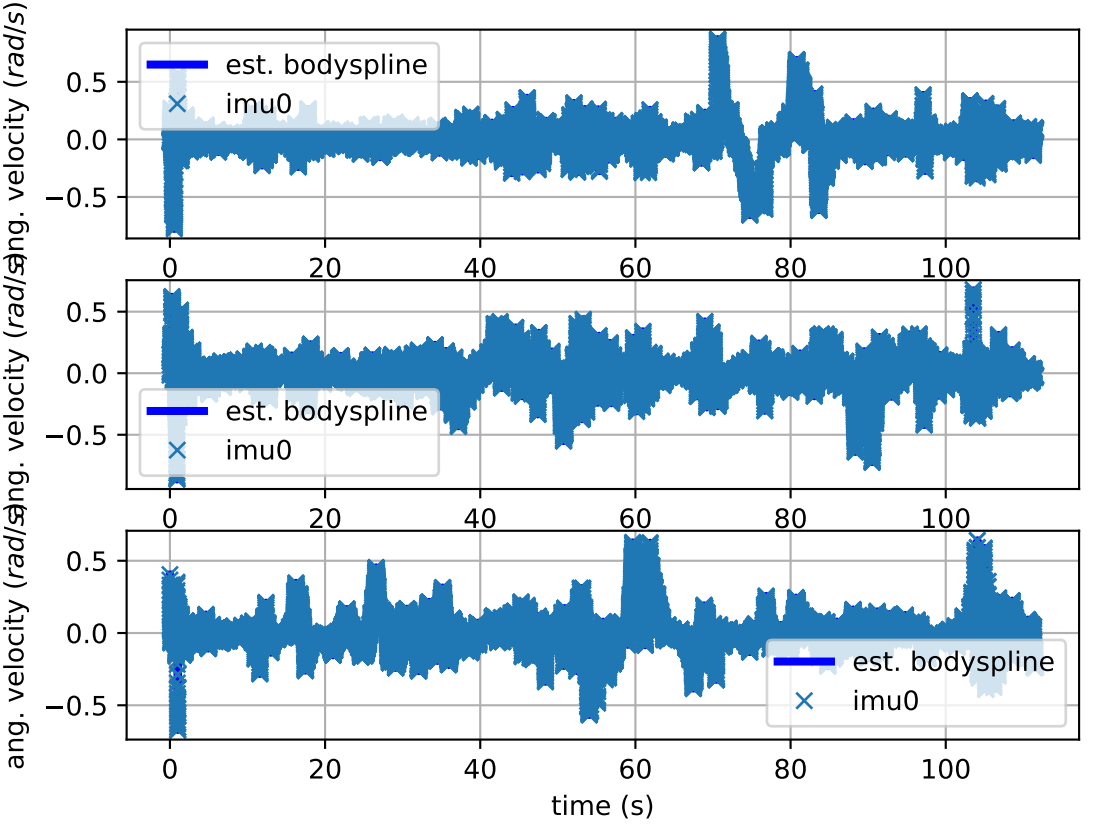

imu0: sample inertial rate
avg dt (ms) = 5.12 +- 0.8891
sample rate (ms)
0
2
4
6
8
20
40
60
80
100
time (s)

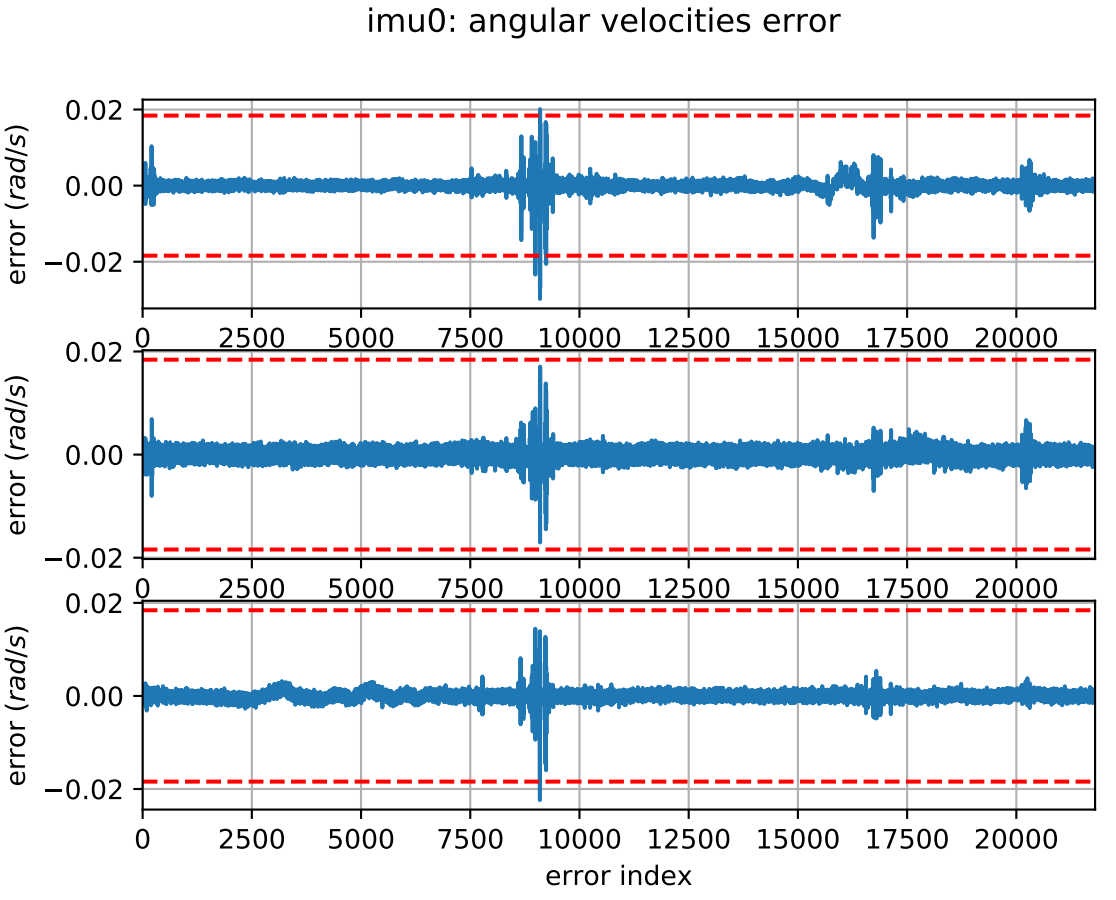

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
est. bodyspline
imu0
5
0
−5
10
8
0
20
40
60
80
100
time (s)

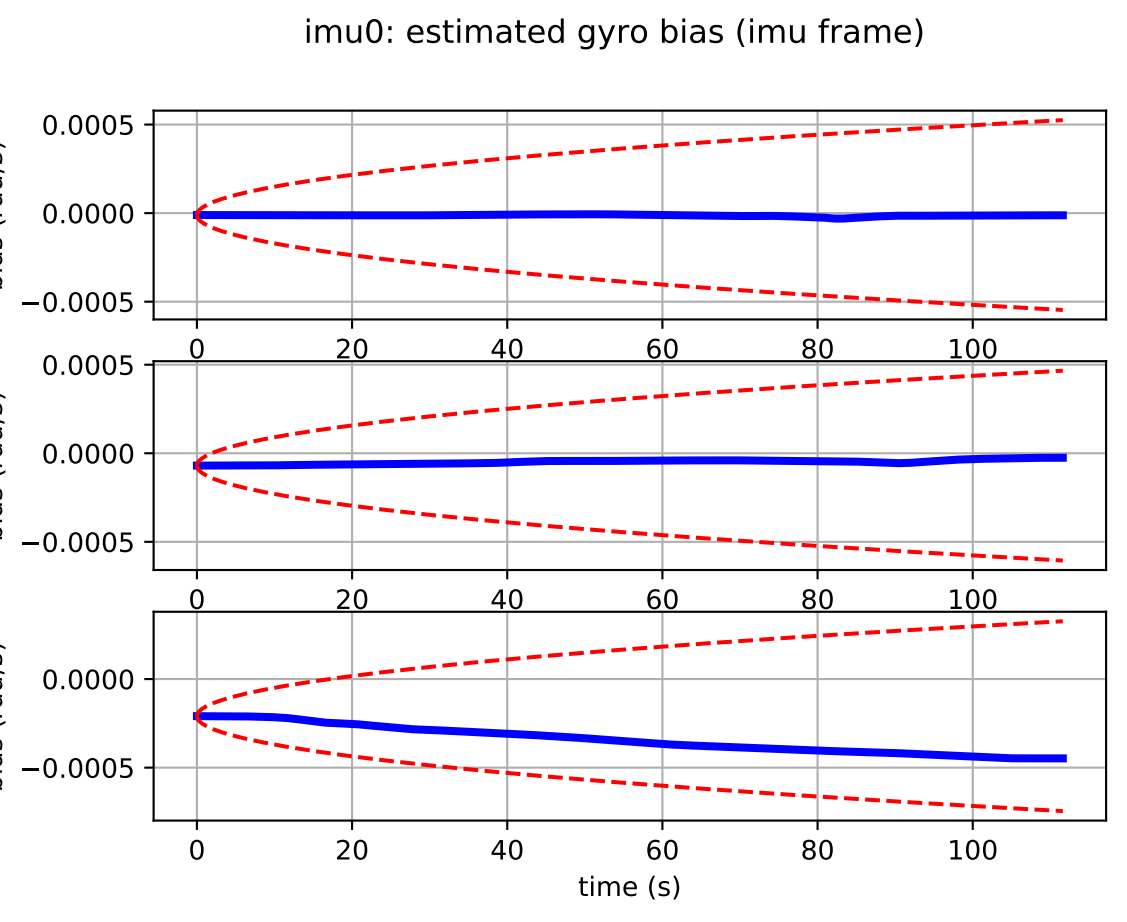

imu0: acceleration error
error (m/s^2)
0.25
0.00
−0.25
0
2500
5000
7500
10000
12500
15000
17500
20000
error index

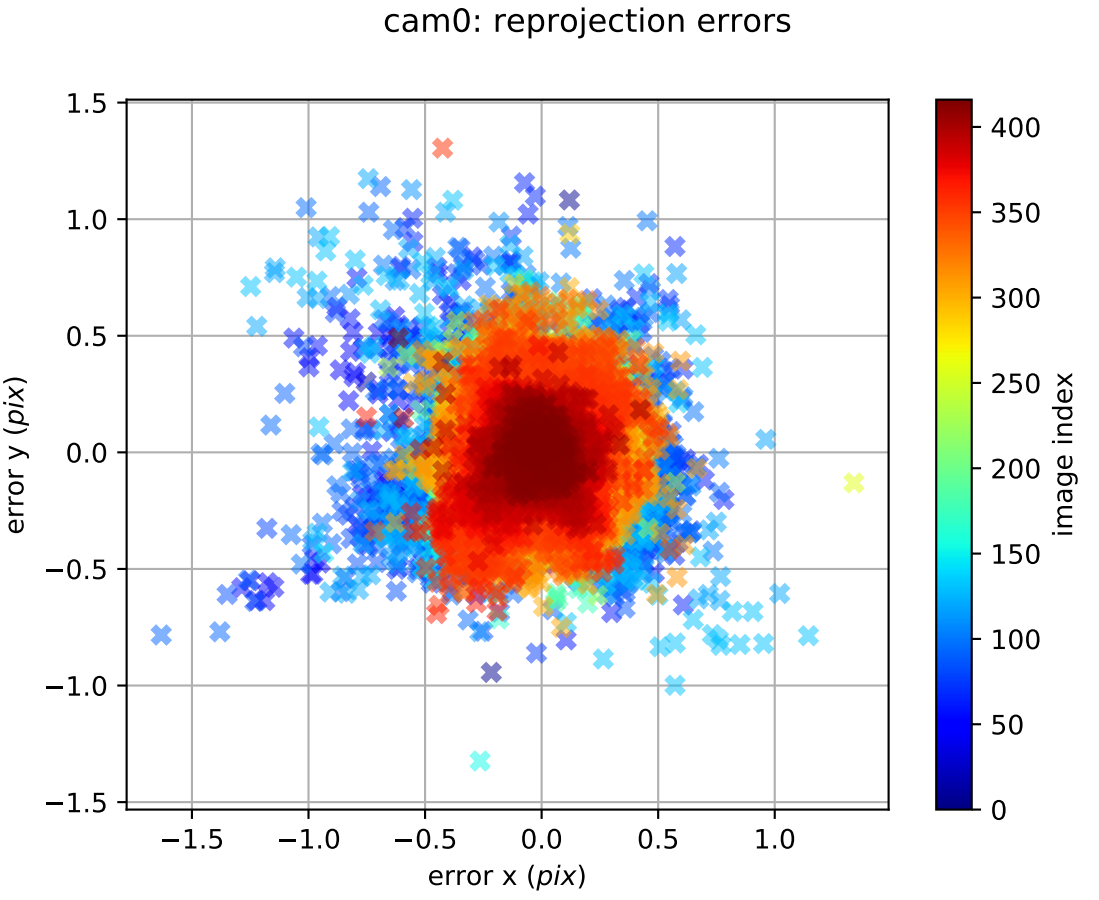

imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.00
−0.02
0
20
40
60
80
100
bias (m/s^2)
−0.04
−0.06
0
20
40
60
80
100
bias (m/s^2)
0.00
−0.02
0
20
40
60
80
100
time (s)

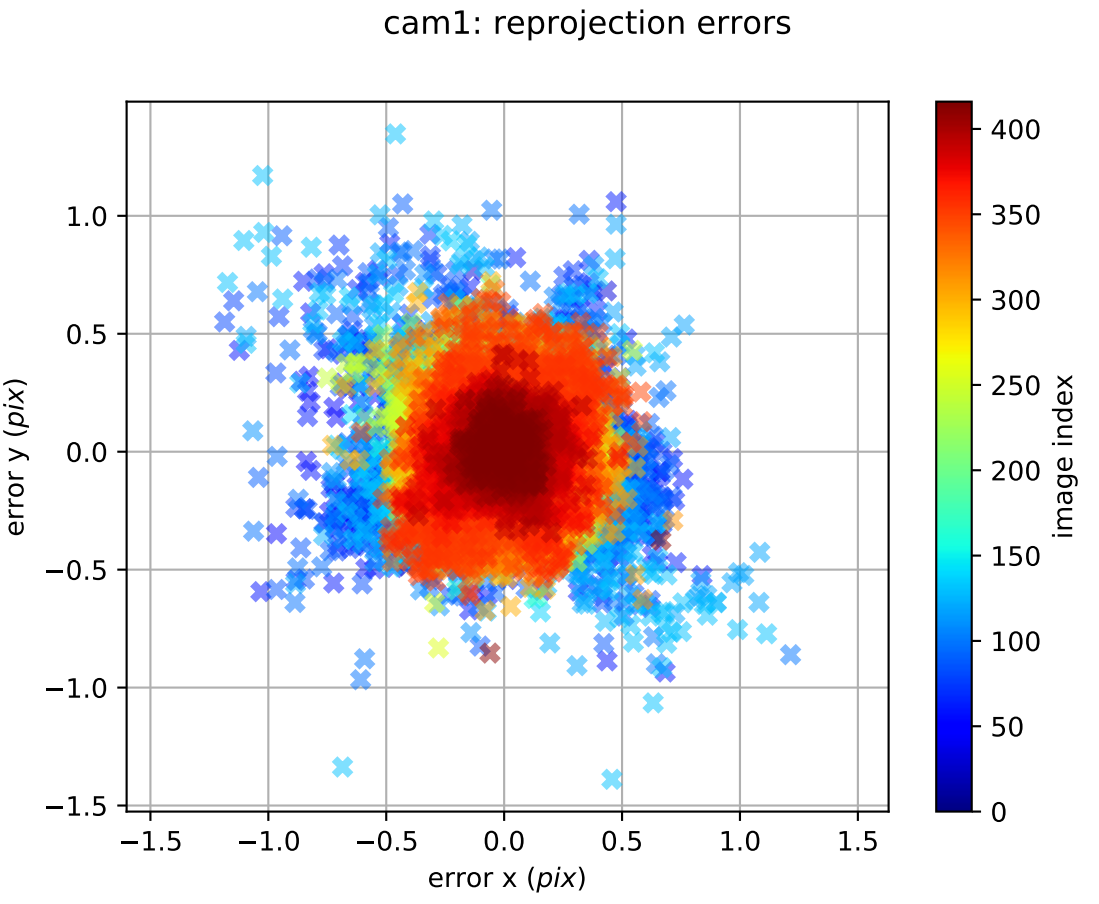
Comparison of predicted and measured angular velocities (body frame)




imu0: angular velocities error
error (rad/s)
0.02
0.00
−0.02
0
2500
5000
7500
10000
12500
15000
17500
20000
error index


imu0: estimated gyro bias (imu frame)
0.0005
0.0000
−0.0005
0
20
40
60
80
100
time (s)


cam0: reprojection errors
error y (pix)
error x (pix)
image index
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.5
−1.0
−0.5
0.0
0.5
1.0
400
350
300
250
200
150
100
50
0


cam1: reprojection errors
error y (pix)
error x (pix)
image index
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
400
350
300
250
200
150
100
50
0